

Chapter 3 project scheduling part 2

❑ Network techniques

- **CPM:** Critical Path Method.
- **PERT:** Program (Project) Evaluation and Review Technique.
- Both, use the same Calculations, almost Similar.
- Identify Critical activities that Cannot be delayed without delaying the Project.
- Estimate the amount of Slack associated with non-critical activities
- Gantt Chart also used in Scheduling.
- Main difference is (PERT) Probabilistic and (CPM) Deterministic in time estimation.

❑ Comparison Between CPM and PERT

CPM (Critical Path Method)	PERT (Program Evaluation and Review Technique)
▪ Uses Network, Calculate float or slack, identify Critical Path and activities, guides to monitor and Controlling Project	▪ Same as CPM
▪ Uses One Value of activity time	▪ Requires 3 estimates of activity time Calculates mean and variance of time
▪ Used where times can be estimated with Confidence, Familiar activities	▪ Used where times cannot be estimated with confidence. ▪ Unfamiliar or New activities
▪ Minimizing Cost is more important	▪ Meeting Time target or estimating Percent Completion is more important
▪ Example: construction projects, building one off machines, ships, etc...	▪ Example: Involving new activities or products, research and development, etc ...

❑ Benefits of CPM and PERT techniques

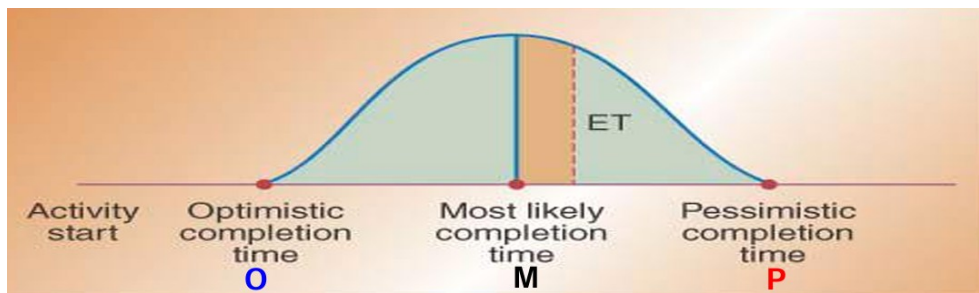
- Shows interdependence of all tasks, work packages, and work units.
- Determines expected project completion date.
- Identifies so-called critical activities, which can delay the project completion time.
- Identified activities with slacks that can be delayed for specified periods without penalty, or from which resources may be temporarily borrowed.
- Shows which activities must be coordinated to avoid resource or timing conflicts.
- Shows which activities may run in parallel to meet project completion date.

❑ PERT (Project Evaluation and Review Technique)

- So far, times can be estimated with relative certainty, confidence.
- For many situations this is Not Possible, e.g., Research, development, new products and projects, etc.
- Use 3-time estimates:
 - O = Optimistic time estimate.
 - M = most likely time estimate, mode.
 - P = Pessimistic time estimate.

❑ Using Beta Probability Distribution to Calculate Expected Time Durations

- A **typical beta distribution** is shown below, note that it has definite end points.
- The expected time for **finishing each activity** is a **weighted average**.



❑ Dealing with Uncertainties

$$\text{Exp. time} = \frac{\text{optimistic} + 4 (\text{most likely}) + \text{pessimistic}}{6}$$

$$\text{Expected Value (TE)} = (O + 4m + P) / 6$$

$$\text{Variance (V)} = ((P - O) / 6)^2$$

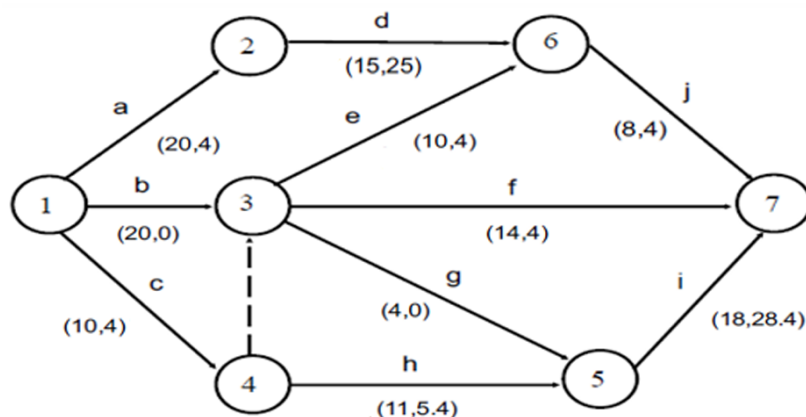
$$\text{Std. Deviation } (\delta) = (\text{SQRT}(V)) = (P - O) / 6$$

❑ Example1:

- Calculating the Probability of Finishing the Project in 50 Days

Activity	Immediate Predecessor	Optimistic Time	Most Likely Time	Pessimistic Time	EXP TE	Var V	S.Dev σ
a	-	10	22	22	20	4	2
b	-	20	20	20	20	0	0
c	-	4	10	16	10	4	2
d	a	2	14	32	15	25	5
e	b,c	8	8	20	10	4	2
f	b,c	8	14	20	14	4	2
g	b,c	4	4	4	4	0	0
h	c	2	12	16	11	5.4	2.32
i	g,h	6	16	38	18	28.4	5.33
j	d,e	2	8	14	8	4	2

❑ Answer



- Assume, PM promised to **complete the project in the 50 days** What are the chances of meeting that deadline? **Calculate Z, where.**

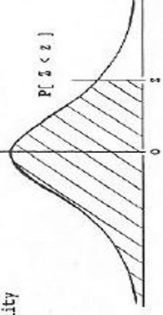
$$z = (D - S) / \sqrt{V}$$

- D = 50;
- S (Scheduled date) = 20+15+8=43;
- V=(4+25+4)=33
- Z = (50 - 43) / 5.745 = 1.22 standard deviations.
- 0.8888 = 89%
- The probability of achieving the scheduled date of day 50 is 89%

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z, i.e.,

$$P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}} dt$$


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9933	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9981	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.1	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.2	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.3	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.4	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.5	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.6	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.7	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.8	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
3.9	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986
4.0	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986	0.9986

Project Crashing

- The Completion Times and Costs of the Project.

Activity	Normal completion time (days)	Shortest completion time (days)	Cost of reduction per day (£)
A (1-2)	6	4	80
B (1-3)	8	4	90
C (1-4)	5	3	30
D (2-4)	3	3	-
E (2-5)	5	3	40
F (3-6)	12	8	200
G (4-6)	8	6	50
H (5-6)	6	6	-

Completion time (days)	Cost (£)
20	9000
17	8790
16	8800
15	8920
13	9180

- The **Minimum Cost for Completing the Project is £8790 in 17-day**, and that the **Shortest Possible Completion Time is 13-day Costing \$9180**.

❑ What is Project Crashing ?

- **Project Crashing** in Project Management - is a Method used to Speed up a Project's Timeline by adding additional resources without changing the Scope of the Project.
- **Project Crashing** - is the Process of accelerating a project.
- **Project Crashing** - also known as Project Time Compression and Crashing the Project Schedule.
- **Reduced Project Completion Time** is "Crashing".

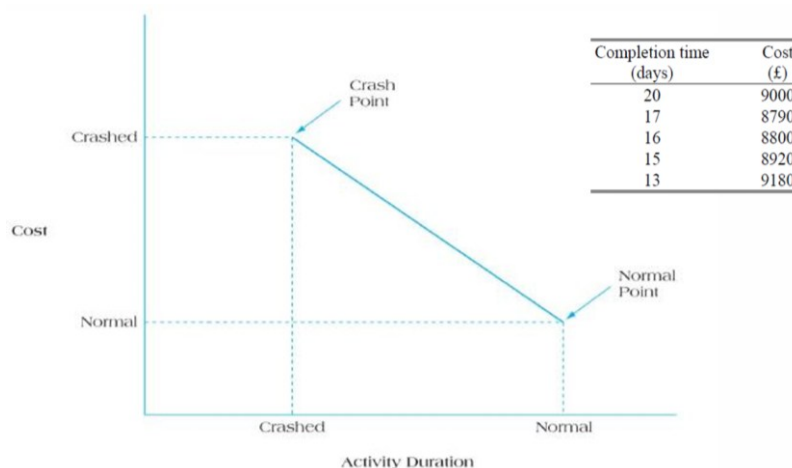
❑ Principal Methods for Crashing

- Improving Existing Resources' Productivity.
- Changing Work Methods.
- Compromise Quality and/or Reduce Project Scope.
- Work Overtime.
- Increasing the Quantity of Resources.

❑ Reducing Project Completion Time

- **Project completion times may need to be Shortened because:**
 - Different deadlines.
 - Penalty clauses.
 - Need to put resources on a new project
 - Promised completion dates.
- **Crashing a project needs to balance.**
 - Shorten a project duration.
 - Cost to shorten the project duration.
- **Crashing a project requires you to know:**
 - Crash time of each activity.
 - Crash cost of each activity.

❑ Time Cost Trade Offs for Crashing Activities



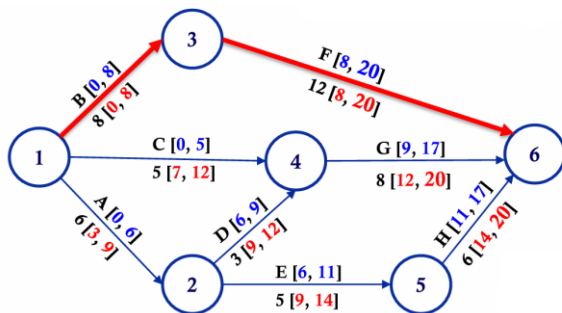
Example 1

Completion of Projects at Minimum Cost:

- A project consisting of 8 activities are described in the following table.
- The Cost for Completion of these 8 activities is £5800 excluding the Site Overhead.
- The Overhead Cost of general site activities is £160/day.
- We are asked to:**
 - Calculate the Normal Completion of the Project, its Cost, and the Critical Path.
 - Calculate and Plot on a graph paper the cost/time function for the project and state:
 - The Minimum Cost and the associated Time.
 - The Shortest Time and the associated Cost.

Activity	Normal completion time (days)	Shortest completion time (days)	Cost of reduction per day (£)
A (1-2)	6	4	80
B (1-3)	8	4	90
C (1-4)	5	3	30
D (2-4)	3	3	-
E (2-5)	5	3	40
F (3-6)	12	8	200
G (4-6)	8	6	50
H (5-6)	6	6	-

Answer



Paths	Time
BF	20
CG	13
ADG	17
AEH	17

The Cost of Project Normal Time = $5800 + (160 \times 20) = 9000$ £

1. Step 1. Reduce Critical Path by 3 days (Reduce activity B by 3 days) =

$$9000 - (160 \times 3) + (90 \times 3) = 8790 \text{ £}$$

2. Step 2. Reduce Critical Path by 1 day (Reduce activities B & A by 1 day) =

$$8790 - (160) + (90 + 80) = 8800 \text{ £}$$

3. Step 3. Reduce Critical Path by 1 day (Reduce activities F & A by 1 day) =

$$8800 - (160) + (200 + 80) = 8920 \text{ £}$$

4. Step 4. Reduce Critical Path by 2 days (Reduce activities F & G & E by 2 day) =

$$8920 - (160 \times 2) + 2(200 + 50 + 40) = 9180 \text{ £}$$

Paths	Time
BF	20 17
CG	13
ADG	17 16
AEH	17 15

Completion time (days)	Cost (£)
20	9000
17	8790
16	8800
15	8920
13	9180

□ The Minimum Cost for Completing the Project is £8790 in 17-day, and that the Shortest Possible Completion Time is 13-day Costing £9180

□ Completion time VS Cost

